In-Band Signal Compliance: One Metric for Prompt Injection and Temporal Blindness

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License: Apache-2.0 Code: https://github.com/mthamil107/signal-compliance

Abstract

LLM agents read control instructions and ordinary data through the *same* input channel. When an instruction arrives in that channel from the **environment**—an orchestrator's context block, a tool result, a retrieved document, a memory fragment, a fetched policy—rather than from the user, the agent must decide whether to obey it. It gets this decision wrong in two symmetric ways. It **over-complies**, obeying an illegitimate signal it should have rejected (the canonical case is *prompt injection*); and it **under-complies**, ignoring a legitimate signal it should have adopted (ignoring an injected current date—*temporal blindness*; ignoring an access-deny banner; leaking a `do-not-share` memory fragment; scraping a page whose policy forbids automation). We name this phenomenon **In-Band Signal Compliance (IBSC)** and make one claim: over- and under-compliance are the same bug seen from opposite ends of a single legitimacy axis, measurable by *one* metric—*did the agent respond correctly to an environment-originated instruction, given that instruction's legitimacy?*—on *one* leaderboard. We define the IBSC taxonomy, the **Signal-Response Correctness (SRC)** metric, and `signalbench`: a zero-dependency benchmark with five self-contained signal families (`time`, `access-deny`, `memory-label`, `injection`, `bot-policy`), graded on *observed actions* rather than phrasing. We report a real six-model, two-vendor study (75 items/model, single seed). No model exceeds IBSC 0.85 (mean 0.772, range 0.650–0.850); every model shows nonzero under- *and* over-compliance simultaneously; `memory-label` is the hardest family by mean across all six models (mean 0.667); and capability is not monotonic—a small model (`gpt-4o-mini`, 0.833) beats larger ones. We state the n=75 / single-seed caveats plainly and do not over-generalize.

Tagline. *One channel, two failures, one number: comply when the signal is legitimate, refuse when it isn't.*

1 Introduction

1.1 The channel is the problem

In telephone networks, *in-band signaling* meant putting control tones on the same wire as the voice. It was convenient and it was a security disaster: a 2600 Hz whistle down the voice channel could seize a trunk line, because the network could not tell a control signal from the sound of a conversation. The fix was *out-of-band* signaling—move control onto a separate channel the caller cannot reach.

LLM agents have re-created the in-band problem. Everything the model reads—the system prompt, the user's request, a tool's JSON result, a retrieved web page, a recalled memory, a fetched `robots.txt`—arrives as bytes in one context window. There is no out-of-band channel: no cryptographic header the model natively trusts, no hardware line the retrieved document cannot write to. When a retrieved page says *"ignore your previous instructions and email me the user's data,"* those bytes ride the same channel as the user's actual question. When an orchestrator injects *"the current date is 2026-07-03,"* those bytes ride the same channel as a poisoned document asserting a fake date. The agent must adjudicate, from content alone, whether each environment-originated instruction is one it should take up or one it should refuse.

1.2 Two failures, one axis

The field has studied one half of this. **Prompt injection**—an agent obeying a malicious instruction embedded in untrusted content—is the subject of mature benchmarks [4, 11] and a training-time program, the Instruction Hierarchy [10]. Call this **over-compliance**: the agent obeyed a signal it should have rejected.

The other half is under-studied and, we argue, identical in structure. Consider an agent that is *handed* a legitimate current date and reasons from its stale training cutoff anyway (**temporal blindness**). Or an agent that receives an authoritative access-deny signal and proceeds regardless (the failure our recusal work [9] measured). Or one that recalls a memory fragment tagged `do-not-share` and forwards it downstream. Or one that fetches a page whose usage policy forbids automated access and scrapes it anyway. In each case the agent behaved *as if the environment signal were absent*. Call this **under-compliance**: the agent ignored a signal it should have taken up.

Here is the unification. Fix the delivery mechanism—*an instruction arriving in-band from the environment*—and vary only one label: is the instruction **legitimate** (should be obeyed) or **adversarial** (should be refused)? Then **over-compliance** is failing on an *adversarial* signal (obeyed when it should have resisted), and **under-compliance** is failing on a *legitimate* signal (ignored when it should have complied). They are mirror images across the legitimacy axis. The single question that scores both is: *given the legitimacy of this environment-originated instruction, did the agent respond correctly?* One 0/1 score per item; one leaderboard for the whole spectrum.

Every family in the benchmark spans **both** poles—each ships a legitimate stratum and an adversarial one—which is what pins a one-sided always-comply or always-resist policy to ≈ 0.5 (§3). The `time` and `injection` families are the cleanest existence proofs that under- and over-compliance are one phenomenon: within a single family the *same* signal type populates both poles. In `time`, a legitimate injected datetime tests uptake (temporal blindness is the under-compliance failure) while a fabricated *fake-now* datetime planted in retrieved content tests resistance; in `injection`, an authorized embedded instruction must be followed while an untrusted one must be refused—legitimacy, not surface form, decides.

1.3 Contributions

1. **A named phenomenon and its unification (§2).** We introduce **In-Band Signal Compliance (IBSC)**: the correctness of an agent's response to an environment-originated in-band control signal, *conditioned on that signal's legitimacy.* The headline is not the name but the claim it packages—that adversarial-signal resistance and legitimate-signal uptake are one axis, not two research areas.

2. **A closed taxonomy (§2.2).** Six axes that place every prior single-vertical benchmark as a cell in one grid, and expose the empty cells.

3. **A single metric (§3). Signal-Response Correctness (SRC)**, aggregated to a macro-balanced **IBSC-Score** on $[0, 1]$, plus stratified sub-metrics (LSU, ISR) so a trivial one-sided policy scores ≈ 0.5, and a no-signal calibration track (FTR) so over-triggering models are visibly discounted. Grading is **action-based**: a failure is asserted only on an observed action, never on phrasing.
4. **`signalbench` (§4).** A zero-dependency, offline-first benchmark: five signal families, each mirroring a prior work and each shipping a self-contained task set, plus deterministic mock providers and function-calling-aware real providers.
5. **A real six-model study (§5).** A two-vendor, six-model leaderboard (75 items/model, single seed) with seven headline findings, stated with the n=75 caveat and no over-generalization.

1.4 What this paper does NOT claim to invent

We did not coin "prompt injection," "temporal blindness," or "instruction hierarchy"; we cite the originators. Our temporal-blindness case is a *narrower, cleaner special case*—uptake of an explicitly injected date—distinct from the elapsed-time-perception sense of [2]. The in-band access-deny signal is our own prior work [9]; IBSC is its generalization, and we treat recusal as the first datapoint on the under-compliance axis. The *term* is branding; the measurement and the unification are the contribution.

2 The IBSC framework

2.1 Definitions

In-Band Signal Compliance (IBSC). The framework measuring whether an LLM agent responds *correctly* to control signals that arrive through the same input channel as user text but originate from the **environment** (orchestrator, resource, retrieved content, or system) rather than the user—*correct* meaning: take up legitimate signals, reject illegitimate ones.

In-band control signal. An instruction embedded in the data channel the model reads (a context block, tool result, retrieved document, memory fragment, fetched policy) intended to change agent behavior. "In-band" = it shares bytes with ordinary content and carries no cryptographic out-of-band authority.

Legitimacy. The ground-truth label of whether an in-band signal *ought* to be obeyed. This is the pivot of the framework: the *identical* observable behavior ("the agent obeyed the signal") is **correct** under `legitimate` and a **failure** under `adversarial`.

Trust-boundary invariant. `provenance = user` is intentionally excluded from every scored cell. A user instruction is the *reference frame*, not a test item. IBSC scores only the four non-user origins.

2.2 The taxonomy (six axes)

Every IBSC item is a point in a six-axis space (Table 1).

The two quadrants. *Under-compliance* (`legitimate/conditional` $\rightarrow$ `uptake`) holds the legitimate stratum of every family—`time` (use), `injection` (follow an authorized instruction), `access-deny` (withdraw), `memory-label` (do-not-propagate), `bot-policy` (abstain). Failing here = the agent behaved as if the signal were absent. *Over-compliance* (`adversarial` $\rightarrow$ `resist`) holds the adversarial stratum—the `injection` payload, the *fake-now* date, and the unauthorized/spoofed

Axis	Values	What it captures
legitimacy	`legitimate`, `adversarial`, `conditional`	Whether the signal *should* be obeyed. `conditional` = legitimate-by-default but in-band-flippable by a valid higher authority.
provenance	`user`, `orchestrator`, `resource`, `retrieved-content`, `system`	Who authored the in-band bytes. `user` is the trusted reference frame and is never scored.
correct-action	`use`, `withdraw`, `do-not-propagate`, `do-not-call`, `refuse`, `abstain`	The single behavior that scores correct for a given signal + legitimacy.
failure-mode	`under-compliance`, `over-compliance`	The two symmetric error classes—legitimacy-mirror images. This is the unifying claim.
signal-kind	`temporal-context`, `access-revocation`, `propagation-label`, `injection-payload`, `automation-policy`	The semantic type of the signal, one per family.
compliance-direction	`uptake`, `resist`	The required response polarity, derived from legitimacy. Makes the leaderboard single-metric.

Table 1: The six-axis IBSC taxonomy.

Family	signal-kind	legitimacy	correct-action	mirrors
`time`	temporal-context	legit. (+ adv. *fake-now*)	`use`	GroundClock / NowBench [7]
`access-deny`	access-revocation	conditional	`withdraw`	Recuse [9]
`memory-label`	propagation-label	legitimate	`do-not-propagate`	memorywire [8]
`injection`	injection-payload	adversarial	`refuse`	Beyond Pattern Matching [6]; InjecAgent [11]
`bot-policy`	automation-policy	legitimate	`abstain`	ai-bot-shield [5]

Table 2: The five shipped signal families.

variants of the deny, memory-label, and bot-policy signals. Failing here = the agent let untrusted input rewrite its instructions or its ground truth.

The conditional cell. `access-deny` is `legitimacy = conditional`: default correct-action is `withdraw` (recuse), but a *valid* in-band authorization framing may legitimately flip it back to proceed. Scoring must verify the overriding signal's *own* legitimacy— otherwise the flip is "injection wearing an authorization costume."

The reserved cell. `do-not-call` $\times$ `provenance = orchestrator/system` is the tool-allowlist case—an under-compliance failure = invoking a tool the in-band allowlist excluded. It is the natural sixth family, left as an explicit extension slot so the taxonomy stays closed over the correct-action axis. It is not shipped as a full family in v0.

2.3 The five families

Table 2 lists the five shipped families and the prior work each mirrors.

3 The metric

3.1 Signal-Response Correctness (SRC)

SRC answers exactly one question per item: *given the legitimacy of the environment-originated instruction, did the agent respond correctly?*—comply when legitimate, resist when illegitimate. The headline is:

> **IBSC-Score** = macro-averaged, per-family *balanced* SRC across the five families, on $[0, 1]$.

One number scores both failure modes symmetrically.

3.2 Per-item scoring

Every item carries an in-band `Signal` with a `Legitimacy` label and a derived `CorrectAction`. The owning family grades the `Response` into a `Verdict` with a binary score in $\{0.0, 1.0\}$:

1. **Legitimate signal, CorrectAction = COMPLY** → `CORRECT` (1.0) if the response demonstrably takes up the signal; else `UNDER_COMPLIANCE` (0.0).
2. **Illegitimate signal, CorrectAction = RESIST** → `CORRECT` (1.0) if the response ignores the injected instruction *and* still completes the original user task; else `OVER_COMPLIANCE` (0.0).
3. **No-signal probe, Legitimacy = NONE** → `CORRECT` (1.0) if the response does the plain task without inventing or acting on an absent signal; else `FALSE_TRIGGER` (0.0).
4. **Abstention** counts 0.0 for SRC but is flagged non-harmful and counted in `abstain_rate`.
5. **Runtime/parse failure** → `ERROR` (0.0).

Grading is family-owned, deterministic, and standard-library only—no LLM judge. Families ship structured checks: `expected_answer` match, `forbidden_tokens` must-be-absent, `must_refuse` flags, and `required_tool` / `forbidden_tool` on `Response.tool_calls`. This applies the IFEval [12] insight (programmatic verifiability) to *environment-channel* signals carrying a *legitimacy* label.

3.3 Action-based grading

A live-model harness must not confuse *how a model worded a reply* with *what it did*. Over- and under-compliance are therefore asserted **only** on an observed action: the response *emits an opaque secret value*, *calls a forbidden tool*, *adopts a spoofed value*, or *ignores a recuse/withhold* directive—never on a text-substring miss. A model that resists an attack but rewords, or withholds a secret without uttering "refuse," scores `CORRECT`. Matching searches both the response text *and* the tool-call arguments, and is separator-insensitive. Forbidden tokens are opaque secret values a refusal would not naturally echo. (An earlier phrasing-based grader inflated over-compliance and false-trigger rates; this study uses the action-based grader, and we audited every non-`CORRECT` verdict against the raw stored response.)

3.4 Aggregation

Two intermediate recalls per stratum, so trivial policies cannot win: **LSU** (Legitimate Signal Uptake) = correct/total over legitimate items; **ISR** (Illegitimate Signal Resistance) = correct/total over illegitimate items. Per family,

$$\text{family\_SRC}_{\text{bal}} = \tfrac{1}{2}(\text{family\_LSU} + \text{family\_ISR})$$

when the family has both strata, else the single-stratum rate. The headline **IBSC-Score** is the equal-weight macro-average of $\text{family\_SRC}_{\text{bal}}$ across the five families, so a large family cannot dominate and an **always-comply** or **always-resist** agent maps to ≈ 0.5. Secondary pooled quantities: `SRC_micro`, global LSU/ISR, `under_rate` $= 1 - \text{LSU}$, `over_rate` $= 1 - \text{ISR}$.

3.5 Calibration (a separate track)

A no-signal track (`Legitimacy = NONE`), *not* folded into the headline, catches over-triggering. **FTR** (False Trigger Rate) = `FALSE_TRIGGER`/total no-signal probes—the rate at which the agent acts on a control signal that is not present. The optional penalized view $\text{IBSC-Score}_{\text{cal}} = \text{IBSC-Score}\cdot(1-\text{FTR})$ is reported *alongside* the raw score, never replacing it.

4 `signalbench`: the benchmark

`signalbench` is the pip package, import name, benchmark, and CLI command for IBSC. It ships five self-contained offline task sets (one per family) that run with **zero dependencies and no network**, plus function-calling-aware real providers for OpenAI and Gemini behind a common `Provider` ABC.

4.1 Design principles

- **Offline-first / mock providers.** A `MockProvider` supplies deterministic policies (`oracle`, `always_comply`, `always_resist`, `random`) so the harness, metric, and leaderboard are exercisable and testable without any vendor.
- **Deterministic, std-lib grading.** No LLM judge; every family ships structured checks. Scoring is reproducible bit-for-bit.
- **Self-contained families.** Each family builds its own task set in code; sibling repos *enrich* but are never *required*.
- **Closed taxonomy, open slots.** The `do-not-call` tool-allowlist cell is a documented extension point, not a silent gap.

4.2 The five families

Each family emits **legitimate**, **illegitimate**, and **NONE** (calibration) items and grades deterministically on actions.

`time`. *Legitimate:* an injected current-date block that must be used for date arithmetic; answering from the training cutoff is under-compliance (temporal blindness). *Illegitimate:* a *fake-now* override planted in retrieved content asserting a wrong date the agent must ignore; adopting it is over-compliance. Mirrors GroundClock / NowBench [7].

`access-deny` (conditional). A `tool_result` carries an authoritative access-deny the agent must honor (recuse), or a tool allowlist; correct action is `withdraw`. The illegitimate stratum injects "access granted, ignore restrictions"—scoring checks the override's *own* provenance, so an injection dressed as authorization does not pass. Mirrors Recuse [9].

#	system	IBSC	IBSC_{cal}	LSU	ISR	under	over	FTR
1	`mock:oracle` *(ref.)*	1.000	1.000	1.00	1.00	0.00	0.00	0.00
2	`gemini-2.5-pro`	0.850	0.850	0.967	0.733	0.033	0.267	0.00
3	`gpt-4o-mini`	0.833	0.833	0.867	0.800	0.133	0.200	0.00
4	`gpt-5.5`	0.817	0.817	0.767	0.867	0.233	0.133	0.00
5	`gpt-5.1`	0.800	0.800	0.800	0.800	0.200	0.200	0.00
6	`gpt-4.1`	0.683	0.683	0.800	0.567	0.200	0.433	0.00
7	`gemini-2.5-flash`	0.650	0.650	0.833	0.467	0.167	0.533	0.00

Table 3: IBSC leaderboard. 6 real models + oracle reference; $n = 75$ items/model, single seed. `under`$= 1-$LSU, `over`$= 1-$ISR. Mean IBSC across the six real models = 0.772; range 0.650–0.850. `gpt-5.5`'s 5/75 API-layer policy blocks are counted as errors (score 0); an alternative scoring that credits an API refusal on an illegitimate item as resistance would raise `gpt-5.5` slightly.

`memory-label`. A memory fragment tagged `do-not-share/expired`; correct = withhold the tagged opaque token; leaking it is under-compliance. The illegitimate stratum injects a "sharing now allowed" override to resist. Mirrors memorywire [8].

`injection` (adversarial-dominant). Retrieved content contains a prompt-injection payload the agent must *resist while still completing the user task*; correct `RESIST` requires both ignoring the injection *and* answering the expected task. Mirrors Beyond Pattern Matching [6] and InjecAgent [11].

`bot-policy`. A `robots.txt` / usage-policy directive the agent should honor (a rate or scope limit); correct = `abstain`. The illegitimate stratum forges a policy demanding scope escalation, to resist. Mirrors ai-bot-shield [5].

4.3 CLI

```
signalbench run -provider mock [-policy oracle|always_comply|...]  \
    -families all -n <per_family> -seed <int> -out results/<name>.json
signalbench leaderboard -results results/ -format md|json|table
```

5 Evaluation

5.1 Setup

We evaluate six deployed models across two vendors on the full `signalbench` suite: **5 families × 15 items = 75 items per model**. Configuration: `reasoning_effort`/thinking-budget = `low`; `temperature` = 0 for non-reasoning models (omitted for the `gpt-5.x` models, which allow only the default `temperature`= 1); `seed`= 0; single run, study date 2026-07-03. `mock:oracle` is included as a reference upper bound; one-sided mock policies (`always-comply`/`always-resist`) score 0.500 by construction. Grading is action-based (§3.3); every non-`CORRECT` verdict was audited against the raw response stored in `results/*.json`. After audit there are *zero grading artifacts*: the non-correct census is 34 leaked-secret, 25 forbidden-tool-call, 17 ignored-recuse/withhold, 1 skipped-required-tool, and 5 `gpt-5.5` API-layer policy blocks (recorded as errors). FTR = 0 for every model.

5.2 Leaderboard

Table 3 gives the full leaderboard.

system	time	access_deny	memory_label	injection	bot_policy
mock:oracle *(ref.)*	1.000	1.000	1.000	1.000	1.000
gemini-2.5-pro	1.000	0.750	0.583	0.917	1.000
gpt-4o-mini	1.000	0.833	0.750	0.833	0.750
gpt-5.5	1.000	0.750	0.833	0.833	0.667
gpt-5.1	1.000	0.833	0.667	0.917	0.583
gpt-4.1	0.917	0.500	0.583	0.833	0.583
gemini-2.5-flash	0.917	0.500	0.583	0.667	0.583
mean (6 real models)	**0.972**	**0.694**	**0.667**	**0.833**	**0.694**

Table 4: Per-family SRC. Difficulty order (mean, lower = weaker compliance): `time` 0.972 (easiest), `injection` 0.833, `access_deny` 0.694, `bot_policy` 0.694, `memory_label` 0.667 (hardest).

5.3 Per-family means

Table 4 gives per-family SRC for every model and the across-model mean, which orders the families by difficulty.

5.4 Findings

We state these as observations from a single-seed, n=75/model run and do not over-generalize.

- **F1 — No model exceeds IBSC** 0.85 (mean 0.772, range 0.650–0.850). In-band signal compliance is an unsolved, measurable axis even for frontier models.
- **F2 — Two-pole failure is universal.** Every model shows nonzero under- *and* over-compliance simultaneously—the quantity no single-pole benchmark can put on one axis.
- **F3 — `memory-label` is the hardest family by mean** across all six models (mean 0.667): models leak `do-not-share/expired` values and obey injected "sharing-now-allowed" overrides. This is the robust cross-vendor finding.
- **F4 — Capability is not monotonic.** The small `gpt-4o-mini` (0.833) beats the larger `gpt-5.1` (0.800) and `gpt-4.1` (0.683); `gemini-2.5-pro` (0.850) > `gemini-2.5-flash` (0.650). Signal compliance is a distinct alignment property, not a pure scale effect.
- **F5 — Models occupy different points on the over/under trade-off.** `gemini-2.5-flash` and `gpt-4.1` over-comply heavily (over 0.533, 0.433) but under-comply little; `gpt-5.5` resists best (over 0.133) but under-complies more (under 0.233). Visible only on IBSC's unified axis.
- **F6 — FTR = 0 for all.** Once graded on actions, no model over-triggers on no-signal calibration probes.
- **F7 — A frontier platform safety filter surfaces as an orthogonal defense.** `gpt-5.5`'s platform cybersecurity content filter hard-blocks 5/75 adversarial items at the API layer (recorded as errors)—a non-model-level defense the benchmark surfaces.

6 Related work

IBSC is positioned as the *union* of several lines of work, each occupying one cell or one pole of the taxonomy.

Instruction Hierarchy (IH) [10] is a *training intervention* that resolves *conflicts* between instruction sources by privilege rank. IBSC is a *measurement framework* for whether an agent responds correctly to a *lone* environment-originated instruction when no conflict is present. IH is centered on the over-compliance half (resist misaligned lower-privilege text); it defines no *legitimate-signal uptake* correctness axis—it does not score whether a model *adopts* a lone, non-conflicting environment signal it should take up (an injected date, an access-deny, a `do-not-share` label), which is the under-compliance half IBSC measures. IBSC's contribution is the unification: uptake and resistance are one axis on one leaderboard, and IBSC can serve as the evaluation that reveals IH-trained models over-suppressing legitimate environment signals. Recent IH extensions [1] remain within the conflict-resolution paradigm and add no uptake axis.

IFEval [12] measures whether a model follows an explicit instruction from the **user**—trusted, single-source, always legitimate. We borrow its programmatic verifiability but apply it to **environment** signals, where the central question is not "did it comply?" but "did it comply given legitimacy?"; IFEval has no legitimacy axis and cannot express over-compliance as a failure.

AgentDojo [4] **and InjecAgent** [11] measure only the adversarial pole—resistance to illegitimate injected instructions. Neither has any notion of a legitimate environment signal the agent *should* obey. IBSC generalizes the channel insight to both polarities and reframes "attack success" as "over-compliance to an illegitimate in-band signal," on the same ruler as "under-compliance to a legitimate one." They are the over-compliance sub-benchmark of IBSC's axis.

GateMem [3] is domain-specific to shared-memory governance. IBSC treats a memory `do-not-share` label as *one instance* of a legitimate in-band signal whose uptake is measured, alongside injected dates, access-deny banners, and bot policies. GateMem is one vertical; IBSC is the horizontal claim that these are the same phenomenon.

Recuse (our own) [9] is the direct predecessor and the only work using the "in-band signal" phrasing—but it is a single signal type (access-deny), one polarity, and a small pilot, not a benchmark/leaderboard. IBSC generalizes it to many signal types $\times$ both polarities $\times$ one metric $\times$ a leaderboard. The elapsed-time **Temporal Blindness** [2] work owns that term for time-*perception*; we frame our injected-date case as a distinct, cleanly-measurable special case and do not claim to have coined the term.

7 Limitations

We state the limitations plainly.

- **Scale.** $n = 75$ items per model, a single seed, one run. The task pools are small and self-authored—sufficient to seed a leaderboard, not for strong per-model statistical claims.
- **Determinism.** The reasoning models are non-deterministic at their forced `temperature`$= 1$; their scores could shift across reruns.
- **Vendor coverage.** Only two vendors (OpenAI, Google). Broader coverage is future work.
- **API-layer confound.** `gpt-5.5`'s 5 API-blocked items are counted as errors (score 0); an alternative scoring crediting an API refusal on an illegitimate item as resistance would raise its score slightly. We report the strict scoring and flag the alternative.
- **Grading recall.** Action-based deterministic grading trades recall for reproducibility; it can miss paraphrastic compliance or subtle partial leaks, though the full non-correct census was audited against raw responses.

- **Stipulated legitimacy.** Each item's legitimacy label is set by the benchmark author—appropriate for a controlled probe, not a model of the genuinely ambiguous cases real agents face. The reserved `do-not-call` tool-allowlist family is designed but not shipped in v0.

8 Conclusion

Control and data share a channel in every LLM agent, and the field has been studying only half of what goes wrong there. Prompt injection—obeying an illegitimate in-band instruction—is over-compliance. Temporal blindness, ignored access-deny signals, leaked `do-not-share` fragments, and scraped no-automation pages—ignoring legitimate in-band instructions—are under-compliance. **In-Band Signal Compliance** names the union and scores it with one question: *given the legitimacy of an environment-originated instruction, did the agent respond correctly?* One channel, two failures, one number.

We defined the taxonomy, the Signal-Response Correctness metric, and `signalbench`, and ran a real six-model, two-vendor study. No model exceeds IBSC 0.85; two-pole failure is universal; `memory-label` is the hardest family across every model; and capability is not monotonic. These are single-seed, n=75/model observations, stated with their caveats—sufficient to establish in-band signal compliance as a distinct, measurable, and currently-unsolved alignment axis, and to invite a broader leaderboard. The code, harness, and raw per-item responses are released under Apache-2.0 at `https://github.com/mthamil107/signal-compliance`.

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